
Coarsened Causal Bayesian Optimization

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Abstract

Causal Bayesian optimization (CBO) finds the intervention that optimizes a target variable, but existing methods assume the full causal DAG over all manipulable variables is known — a level of structural knowledge practitioners rarely have. We introduce Coarsened CBO (CCBO), which requires only a *partition* of the manipulable variables into clusters together with the cluster-level graph (a C-DAG): identification, exploration-set construction, and the causal GP prior are all derived from the quotient structure. We prove that this confers structural robustness — *any* misspecification of edges inside a cluster is provably invisible to CCBO — and we characterize the price of coarsening: the attainable optimum degrades monotonically along the coarsening lattice, with equality exactly when an optimal intervention set is a union of clusters. On two synthetic benchmarks and a light-tunnel benchmark built on a physical simulator, CCBO with the correct fineness matches full-knowledge CBO, coarse partitions trade optimality for robustness exactly as predicted, and intra-cluster misspecifications that structurally damage fine-grained CBO (removing its best intervention from the exploration set) leave CCBO’s trajectories byte-for-byte unchanged.

1 Introduction

Causal Bayesian optimization [Aglietti et al., 2020] unifies two ideas: when the variable Y we wish to optimize sits inside a causal system, the right search space is the space of *interventions* ($X_S = \mathbf{x}_S$), and observational data plus the causal graph yield informative priors over the interventional response surfaces. The catch is the input requirement. CBO and its descendants [Aglietti et al., 2021, 2023, Sussex et al., 2023] assume the full DAG over all observables — every edge, every latent confounder — is specified correctly. Practitioners typically have something weaker and more reliable: they know the *groups*. An experimentalist driving a light tunnel knows that the LED color channels form one subsystem and the polarizers another, without committing to the wiring *within* the color subsystem; a reef ecologist distinguishes nutrient-related stressors from temperature-related ones without specifying edges among the former. The graph-free end of the spectrum is served by methods that learn structure online [Mukherjee et al., 2024, Branchini et al., 2023] at a heavy per-trial cost. The middle of the spectrum — cluster-level knowledge — has well-developed identification theory [Anand et al., 2023, Ferreira & Assaad, 2025, Wahl et al., 2024] but, to our knowledge, no optimization method.

CCBO fills this gap. The practitioner supplies a partition π of the manipulable variables; CCBO builds the quotient graph $\mathcal{G}^\pi$ (a C-DAG over clusters, with bidirected edges absorbing latent confounding via the latent projection of Lee et al., 2019), restricts exploration to cluster-level interventions whose effects are identifiable from $\mathcal{G}^\pi$, and uses the identified cluster-level do-calculus estimates as GP prior means. The central structural insight is that this construction never looks inside a cluster:

Contributions.

- We formalize CCBO: Bayesian optimization over cluster-level interventions with identification performed on the C-DAG by a complete algorithm [Shpitser & Pearl, 2006], an identifiability-gated exploration set, and a two-tier causal GP prior (§3).
- We prove an *intra-cluster invariance* property (Prop. 2): every quantity CCBO computes is invariant to arbitrary edge misspecification within clusters — and a *price-of-coarsening* characterization (Prop. 4): the attainable optimum is monotone along the coarsening lattice, with an exact condition for losslessness. We also show, contrary to what one might expect, that refining a valid partition can break validity (Remark 3), while exploration and identifiability are monotone under valid refinements (Prop. 3).
- We evaluate on CompleteGraph and SimplifiedCoralGraph [Aglietti et al., 2020] and on a light-tunnel benchmark built on the causalchamber physical simulator [Gamella et al., 2025], separating two damage regimes of DAG misspecification — prior bias (recoverable by the BO loop) and exploration-set damage (unrecoverable) — and verifying that coarsening is provably immune to both when the misspecification is intra-cluster (§6).

2 Background

SCMs and interventions. A structural causal model $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ induces a distribution over observables $\mathbf{V}$ and, for any $S \subseteq \mathbf{V}$, interventional distributions $P(\cdot \mid (X_S = \mathbf{x}_S))$ obtained by replacing the mechanisms of X_S with constants [Pearl, 2009]. The induced graph $\mathcal{G}$ has a directed edge per functional argument and a bidirected edge per shared latent; projecting out latents yields an ADMG over $\mathbf{V}$ [Lee et al., 2019].

Causal Bayesian optimization. Given manipulable variables $\mathbf{X} \subseteq \mathbf{V}$ and target Y , CBO [Aglietti et al., 2020] solves

$$(S^*, \mathbf{x}^*) = \arg \min_{\substack{S \in \mathfrak{S} \\ \mathbf{x}_S \in \mathcal{X}_S}} \mathbb{E}[Y \mid (X_S = \mathbf{x}_S)], \quad (1)$$

maintaining one GP per intervention set S in an exploration set $\mathfrak{S}$, with prior mean given by the do-calculus estimate of $\mathbb{E}[Y \mid (X_S = \mathbf{x}_S)]$ from observational data and the assumed graph, and selecting $(S, \mathbf{x})$ by expected improvement [Garnett, 2023]. The exploration set is pruned structurally (MIS/POMIS; Lee & Bareinboim, 2018); both the pruning and the priors are functions of the assumed graph, which is where misspecification enters.

Coarsening and C-DAGs.

Definition 1 (Coarsening [Madaleno et al., 2026]). *Given a DAG $\mathcal{G}$ on $\mathbf{V}$, a **coarsening** is a partition $\pi = \{C_1, \dots, C_{d'}\}$ of $\mathbf{V}$ such that the induced quotient $\mathcal{G}^\pi$ — nodes $\{C_1, \dots, C_{d'}\}$, edge $C_k \rightarrow C_\ell$ ($k \neq \ell$) iff $\exists V_i \in C_k, V_j \in C_\ell$ with $V_i \rightarrow V_j$ in $\mathcal{G}$ — is acyclic. π is then called valid.*

For ADMGs the quotient additionally carries $C_k \leftrightarrow C_\ell$ iff some $V_i \in C_k, V_j \in C_\ell$ are connected by a bidirected edge; intra-cluster bidirected edges are dropped. The resulting object is a C-DAG in the sense of Anand et al. [2023], who prove that do-calculus on the C-DAG is sound and complete for the equivalence class of fine-grained graphs that quotient to it. Markov properties transfer to the cluster level, but faithfulness does not [Wahl et al., 2024].

3 Coarsened Causal Bayesian Optimization

3.1 Setup

The practitioner partitions only the manipulable variables: π is a partition of $\mathbf{X}$, the target Y is kept as a singleton, and every non-manipulable observable remains an atomic (singleton) vertex of the quotient. Mixing manipulable and non-manipulable variables in one cluster is disallowed — intervening on a cluster must be a physically executable action. Hidden confounders are first projected out of $\mathcal{G}$ [Lee et al., 2019], giving an ADMG $\mathcal{G}^*$ over observables; the coarsened graph $\mathcal{G}^\pi$ is the quotient of $\mathcal{G}^*$ under π (plus atomic vertices), and validity (acyclicity of the directed part) is checked. Intervening on a cluster C_k means jointly setting all of its members: $(C_k = \mathbf{c}_k)$. The

Algorithm 1 CCBO: Coarsened Causal Bayesian Optimization

Require: Assumed graph $\mathcal{G}$ (or directly $\mathcal{G}^\pi$); partition π of $\mathbf{X}$; target Y ; budget T ; observational data $\mathcal{D}_{\text{obs}}$; cluster budget m .

- 1: $\mathcal{G}^* \leftarrow$ latent projection of $\mathcal{G}$; $\mathcal{G}^\pi \leftarrow$ quotient of $\mathcal{G}^*$ under π ; check acyclicity.
- 2: Build $\mathfrak{S}^\pi$ via Eqs. (2), (3) (one complete-ID call per candidate).
- 3: For each $S \in \mathfrak{S}^\pi$: fit the prior μ_0^S via Eq. (4) on $\mathcal{D}_{\text{obs}}$.
- 4: Collect initial interventional data $\mathcal{D}_0$.
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Update GP posteriors (μ_t^S, k_t^S) for all $S \in \mathfrak{S}^\pi$.
- 7: $(S_t, \mathbf{x}_t) \leftarrow \arg \max_{S, \mathbf{x}} \text{EI}_{S,t}(\mathbf{x}) / \text{cost}(S, \mathbf{x})$.
- 8: Intervene, observe y_t ; update $\mathcal{D}_t$.
- 9: **end for**
- 10: **return** $(S_{t^*}, \mathbf{x}_{t^*})$ where $t^* = \arg \min_t y_t$.

candidate exploration set contains all unions of at most m clusters:

$$\mathfrak{S}_0^\pi = \left\{ \bigcup_{C_k \in \mathcal{S}} C_k \mid \emptyset \neq \mathcal{S} \subseteq \pi, |\mathcal{S}| \leq m \right\}. \quad (2)$$

Remark 1 (Intervention granularity). *Merging manipulable variables coarsens the action space: CCBO can only set a cluster jointly, never a strict subset of it. This is the mechanism behind both the robustness (§4) and the price (Prop. 4) of coarsening.*

3.2 Identifiability-gated exploration and the two-tier prior

For each candidate $S \in \mathfrak{S}_0^\pi$ we pose the cluster-level query $P(Y \mid (S))$ on $\mathcal{G}^\pi$ and run the complete ID algorithm of Shpitser & Pearl [2006] (via ananke; Lee et al., 2023). Following the arm-pruning tradition of structural causal bandits [Lee & Bareinboim, 2018], CCBO *commits to its structural beliefs*: the exploration set $\mathfrak{S}^\pi$ keeps exactly the candidates whose effect is identifiable from $\mathcal{G}^\pi$,

$$\mathfrak{S}^\pi = \{ S \in \mathfrak{S}_0^\pi : P(Y \mid (S)) \text{ identifiable from } \mathcal{G}^\pi \}. \quad (3)$$

This gate is exact (ID is complete on the C-DAG, and identification from $\mathcal{G}^\pi$ is sound for every fine graph in its class; Anand et al., 2023), and it is what transmits structural misspecification into behaviour: a wrong assumed graph keeps or removes *arms*, not just priors (§6.3). The alternative design — exploring non-identified arms under an uninformative prior — trades this structural damage for exploration overhead; we discuss it in §8.

For $S \in \mathfrak{S}^\pi$, the GP prior mean is constructed in two tiers. The ID gate guarantees a non-parametric formula exists; our *estimator* instantiates it as a cascade of GP-based plug-ins (backdoor adjustment $\rightarrow$ front-door $\rightarrow$ g -computation) on the bidirected-edge expansion of $\mathcal{G}^\pi$. The cascade is sound but not complete, so on the rare queries it cannot instantiate we fall back to the observational mean:

$$\mu_0^S(\mathbf{x}_S) = \begin{cases} \hat{\mathbb{E}}[Y \mid (X_S = \mathbf{x}_S)] & \text{estimator cascade instantiates the ID formula,} \\ \bar{Y}_{\text{obs}} & \text{otherwise (logged as a diagnostic).} \end{cases} \quad (4)$$

Because clusters are manipulable-only and interventions assign full clusters, the cluster-level formulas expand exactly into fine-grained regressions on observational data — no sub-cluster heuristic is involved.

Assumption 1 (Valid partition). *The practitioner supplies a partition π of $\mathbf{X}$ whose quotient $\mathcal{G}^\pi$ is acyclic. (Validity is checkable from cluster-level knowledge alone and is verified by the implementation.)*

Assumption 2 (Estimator-completeness boundary). *The ID gate (Eq. 3) is complete; the GP estimator cascade behind Eq. (4) is sound but not complete. Queries that pass the gate but miss the cascade are reported and receive the uninformative tier. In all benchmarks of §6 this set is empty.*

3.3 Algorithm

4 Theoretical properties

Throughout, π, π' are manipulable-only partitions as in §3.1; $\pi_\top$ denotes the identity (all-singleton) partition; π' *refines* π ($\pi' \preceq \pi$) iff every cluster of π' is contained in a cluster of π .

Proposition 1 (Identity recovery). *Under $\pi = \pi_\top$, $\mathcal{G}^\pi = \mathcal{G}^*$ and $\mathfrak{S}_0^\pi$ is the set of all subsets of $\mathbf{X}$ of size at most m ; CCBO reduces to CBO with an identifiability-gated exploration set.*

Proof. Singleton clusters make the quotient map the identity on $\mathcal{G}^*$, and unions of at most m singletons are exactly the subsets of size at most m . $\square$

Proposition 2 (Invariance to intra-cluster misspecification). *Let π be valid for $\mathcal{G}$ and let $\mathcal{G}'$ differ from $\mathcal{G}$ by an arbitrary set of insertions, deletions, or reversals of edges (directed or bidirected) whose endpoints lie within a single cluster of π , with $\mathcal{G}'$ acyclic. Then $(\mathcal{G}')^\pi = \mathcal{G}^\pi$, and consequently the exploration set (Eq. 3), every identification formula, every prior (Eq. 4), and — given the same data and seeds — the entire CCBO trajectory are identical under $\mathcal{G}$ and $\mathcal{G}'$.*

Proof. The latent projection removes only latent vertices; an edge between two observables is never an interior vertex of a projection path, so $(\mathcal{G}')^*$ and $\mathcal{G}^*$ differ exactly by the modified intra-cluster edges. The quotient keeps a directed (resp. bidirected) cluster-level edge iff some fine edge crosses two *distinct* clusters; edges within a cluster contribute nothing. Hence the quotients coincide. All remaining objects — $\mathfrak{S}^\pi$, ID formulas, priors — are functions of $\mathcal{G}^\pi$ and data alone. $\square$

Remark 2 (What invariance buys). *Prop. 2 is a statement about the method, not about estimation error: CCBO cannot even represent an intra-cluster hypothesis, so no intra-cluster mistake can affect it. §6.3 shows the same mistakes can structurally damage fine-grained CBO by deleting its best arm.*

Remark 3 (Refinement does not preserve validity). *One might hope that refining a valid partition is always safe. It is not: with $\mathbf{X} = \{a_1, a_2, b_1, b_2\}$ and fine edges $a_1 \rightarrow b_1, b_2 \rightarrow a_2$, the coarsest partition $\{\{a_1, a_2, b_1, b_2\}\}$ is valid (both edges are intra-cluster) while its refinement $\{\{a_1, a_2\}, \{b_1, b_2\}\}$ induces the 2-cycle $A \rightleftharpoons B$. Validity must therefore be re-checked after every refinement.*

Proposition 3 (Monotonicity under valid refinement). *Let $\pi' \preceq \pi$ with both valid, and take $m = \infty$ in Eq. (2). Then (i) $\mathfrak{S}_0^\pi \subseteq \mathfrak{S}_0^{\pi'}$; (ii) every $S \in \mathfrak{S}^\pi$ is also in $\mathfrak{S}^{\pi'}$, i.e. effects identifiable from $\mathcal{G}^\pi$ remain identifiable from $\mathcal{G}^{\pi'}$, with the same functional.*

Proof. (i) A union of π -clusters is a union of π' -clusters because each π -cluster is. (ii) Quotienting is compositional: $\mathcal{G}^\pi$ is the quotient of $\mathcal{G}^{\pi'}$ under the induced partition of π' -clusters, so every fine graph compatible with $\mathcal{G}^{\pi'}$ is compatible with $\mathcal{G}^\pi$. Identifiability from $\mathcal{G}^\pi$ means a single functional of the observational distribution equals $\mathbb{E}[Y \mid (S)]$ across all SCMs whose graph is compatible with $\mathcal{G}^\pi$ [Anand et al., 2023]; a fortiori the same functional works across the smaller class compatible with $\mathcal{G}^{\pi'}$. $\square$

Proposition 4 (Price of coarsening). *Let $\mu^*(S) = \min_{\mathbf{x}_S} \mathbb{E}[Y \mid (X_S = \mathbf{x}_S)]$ and $V(\pi) = \min_{S \in \mathfrak{S}_0^\pi} \mu^*(S)$, with $m = \infty$. For any valid $\pi' \preceq \pi$:*

$$V(\pi_\top) \leq V(\pi') \leq V(\pi),$$

and $V(\pi) = V(\pi_\top)$ iff some optimizer S^ of Eq. (1) is a union of π -clusters.*

Proof. By Prop. 3(i), $\mathfrak{S}_0^\pi \subseteq \mathfrak{S}_0^{\pi'} \subseteq \mathfrak{S}_0^{\pi_\top} = 2^{\mathbf{X}} \setminus \{\emptyset\}$; minima over larger sets are no larger. Equality holds iff the unrestricted argmin intersects $\mathfrak{S}_0^\pi$, which is exactly the stated condition. $\square$

Remark 4 (Damage taxonomy for misspecification). *Under the gated design (Eq. 3), a wrong assumed graph can hurt in two distinct ways. Tier-1 (prior bias): the arm survives but its prior mean is wrong; expected improvement wastes evaluations, and the interventional data eventually correct it — the damage is a convergence-speed cost. Tier-2 (arm damage): the wrong graph flips the identifiability verdict, deleting an arm from $\mathfrak{S}^\pi$ (or admitting a spurious one); if the deleted arm contained the optimum, no amount of data recovers it. Prop. 2 says coarsening is immune to both tiers whenever the misspecified edges are intra-cluster.*

Table 1: Positioning of CCBO against the nearest methods.

Method	What is learned	Structural prior assumed	Search space
CBO [Aglietti et al., 2020]	nothing (oracle DAG)	full DAG	MIS
GACBO [Mukherjee et al., 2024]	posterior over DAGs	DAG family	union of MIS
Durand et al. 2025	$\text{Pa}(Y) \subseteq \mathbf{X}$	Y leaf	$\mathbf{X}$
Park et al. 2025 [Park et al., 2025]	$\text{PAG} \rightarrow \text{MAG}$	MEC	partial $\mathbf{X}$
SCB [Lee & Bareinboim, 2018]	nothing (oracle DAG)	full DAG	POMIS (atomic)
CCBO (ours)	nothing	partition of $\mathbf{X}$ + C-DAG	ID-gated cluster unions

5 Related work

Methods that relax CBO’s full-graph requirement either learn structure online — GACBO maintains a posterior over plausible DAGs at substantial per-trial cost [Mukherjee et al., 2024], local-discovery CBO learns around Y [Branchini et al., 2023], Durand et al. (2025) identify $\text{Pa}(Y)$ under a leaf assumption — or assume an equivalence class is known [Park et al., 2025]. CCBO instead assumes *less structure of a different kind*: a partition with cluster-level edges, learning no edges at all. The identification side of this regime is established — C-DAG do-calculus [Anand et al., 2023], its extension to C-DMGs over ADMGs [Ferreira & Assaad, 2025], cluster-level Markov properties [Wahl et al., 2024], and recent work on relaxing partition admissibility [Tikka et al., 2025] and on C-DAGs as background knowledge for discovery [Wang et al., 2025] — but none of these perform optimization.

Multi-fidelity and abstraction-based BO. Closest in spirit is the concurrent position paper of Zeitler [2025], which proposes MFACBO — a hybrid of CBO and multi-information-source BO [Poloczek et al., 2017] unified through approximate causal abstraction [Beckers & Halpern, 2019]. MFACBO coarsens the *outcome* (Y_{low} as a cheap, biased proxy of Y_{high}); CCBO is complementary in that it coarsens the *manipulable inputs* into cluster-level interventions and exploits identification on the resulting C-DAG. The two axes are orthogonal and could be combined in a single surrogate; we leave the hybrid to future work.

6 Experiments

6.1 Setup

Benchmarks. (1) *CompleteGraph* [Aglietti et al., 2020]: six observables, latents $U_1 \rightarrow \{A, Y\}$, $U_2 \rightarrow \{B, Y\}$, manipulable $\mathbf{X} = \{B, D, E\}$; the projected ADMG carries $A \leftrightarrow Y$ and $B \leftrightarrow Y$. Exactly two manipulable-only coarsenings are valid: the identity and $\{B\}, \{D, E\}$ (merges containing B with D create cycles through C). (2) *SimplifiedCoralGraph* [Aglietti et al., 2020]: eleven observables, manipulable $\mathbf{X} = \{N, O, C, T, D\}$, 47 valid coarsenings. (3) *LightTunnel*: a benchmark we built on the deterministic light-tunnel simulator of `causalchamber` [Gamella et al., 2025]: manipulable LED channels $\{R, G, B\}$ and polarizer angles $\{P_1, P_2\}$, target Y the (scaled) vis_3 sensor reading, and a latent $U_{\text{color}} \rightarrow \{B, G\}$ giving observational $\text{corr}(B, G) \approx 0.67$. Polarizer interventions are range-limited (hardware-limit style), so the informative structure lies in the color channels.

Protocol. All methods on a benchmark share the same seeds (0–9), the same number of initial interventional points, the same trial budget ($T = 40$), unit costs, and the same per-benchmark cap m on jointly intervened clusters; a fairness check enforces identical trajectory lengths and seed coverage in every results file. Methods: BO (non-causal, full $|\mathbf{X}|$ -dimensional search), CBO on the true fine-grained projected ADMG, GACBO [Mukherjee et al., 2024] with a plausible family containing the true graph and its misspecified variants, and CCBO under representative partitions.

6.2 Main comparison and the coarsening sweep

Table 2 and Figure 1 give the main comparison. To interpret it we computed the ground-truth value of every arm on *CompleteGraph* by brute-force Monte Carlo on the true SEM: the unique

Table 2: Final best Y (mean $\pm$ s.e., 10 seeds $\times$ 40 trials) per method. Tables are auto-generated from the result files; both included files end with their own \bottomrule.

CompleteGraph		SimplifiedCoralGraph	
Method	Final Y	Method	Final Y
BO	-0.628 ± 0.001	BO	9282.608 ± 5.136
CBO	-3.108 ± 0.195	CBO	36.100 ± 0.027
GACBO	-3.204 ± 0.182	GACBO	36.064 ± 0.013
CCBO-finest	-3.502 ± 0.029	CCBO-all-merged	9304.602 ± 3.979
CCBO- $\{B\}, \{DE\}$	-1.245 ± 0.065	CCBO-finest	36.064 ± 0.014
		CCBO- $\{CNO\}, \{DT\}$	36.486 ± 0.018

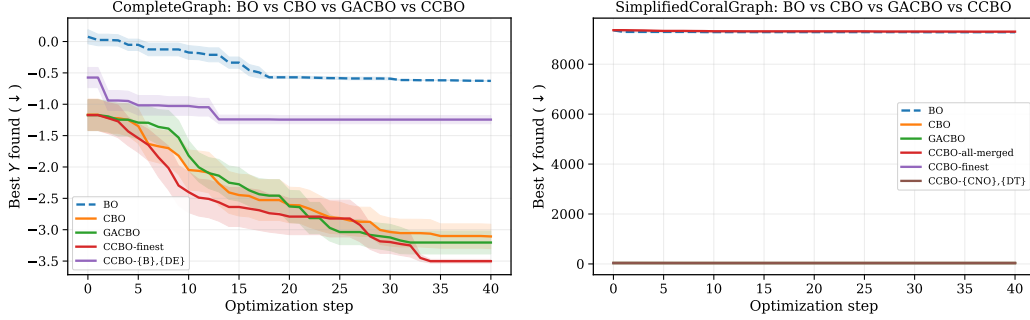


Figure 1: Convergence (mean $\pm$ s.e., 10 seeds): CompleteGraph (left) and SimplifiedCoralGraph (right, log-scaled gap between the BO-like and CBO-like regimes).

optimum is the *joint* arm $\{B, D\}$ with $\mu^*(\{B, D\}) = -3.60$ ($B = -5$, $D = 3.3$), and — a textbook instance of the “where to intervene” phenomenon of Lee & Bareinboim [2018] — additionally forcing E destroys it ($\mu^*(\{B, D, E\}) = -0.57$): left alone, $E = f(A, \cdot)$ inherits U_1 ’s value through A and partially compensates U_1 ’s direct effect on Y . Every entry of Table 2 now reads as “predicted floor $V(\pi)$, minus a small budget gap”: CCBO-finest attains -3.50 ± 0.03 against $V(\pi_\top) = -3.60$; the coarse partition $\{B\}, \{D, E\}$ attains -1.245 ± 0.065 against its Prop. 4 floor $V(\pi) = \mu^*(\{B\}) = -1.31$ — the optimal action (B, D) requires splitting the $\{D, E\}$ cluster and is simply not representable; BO attains -0.63 against $\mu^*(\{B, D, E\}) = -0.57$ for its single joint arm.

CCBO-finest and the CBO baseline deserve a careful reading. By Prop. 1 they are the *same algorithm*; both exploration sets contain the optimal arm $\{B, D\}$, and both use the same adjustment sets (e.g. $\mathbf{Z} = \{A, C\}$ for (E)). The gap (-3.50 ± 0.03 vs. -3.11 ± 0.20) is an implementation-quality statement about the priors — the original release’s hand-coded do-functions use fixed GP hyperparameters and intermittently trap a seed short of the 2-D optimum within the 40-trial budget, whereas the complete-ID pipeline’s estimators are calibrated per query — not evidence that coarsening helps at $\pi_\top$. GACBO sits in between (-3.20 ± 0.18); on SimplifiedCoralGraph all full-knowledge methods agree to three digits (36.06–36.10). The two extremes bracket the spectrum: plain BO and the all-merged partition both collapse to $\approx 9.3 \times 10^3$ on SimplifiedCoralGraph — merging everything hands CCBO exactly BO’s joint $|\mathbf{X}|$ -dimensional search problem, causal prior notwithstanding.

The coarsening sweep (Figure 2) runs CCBO across the valid lattice (all coarsenings for CompleteGraph; a structured finest-to-coarsest sample of the 47 for SimplifiedCoralGraph) and traces the $V(\pi)$ staircase of Prop. 4 empirically. The price is graded and merge-specific, not a function of fineness alone: merging $\{C, D\}$ is *lossless* (36.05, indistinguishable from the finest 36.06 — no optimal set separates C from D), $\{C, N\}$ costs little (36.45), $\{C, D, O\}$ costs 40% (51.2), and $\{C, D, N, O\}$ is catastrophic (4.3×10^3). Which clusters you merge matters far more than how many; Prop. 4 states the exact losslessness condition.

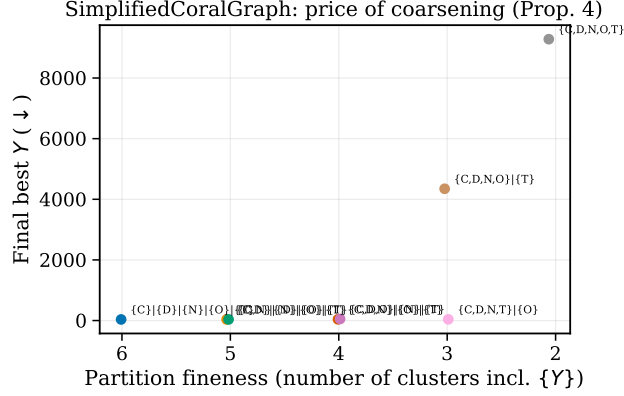


Figure 2: Price of coarsening on SimplifiedCoralGraph: final Y (mean $\pm$ s.e., 10 seeds) across a finest-to-coarsest sample of the valid-partition lattice. The staircase of Prop. 4 is graded and merge-specific.

Table 3: Tier-1 robustness (CompleteGraph; 10 seeds): prior-bias damage is partially self-healing for the fine partition and absent for the coarse one.

Partition	Assumed DAG	$ \mathcal{G}^\pi $	Final Y	Gap vs. correct
finest	NoBC	7	-3.239 ± 0.199	$+0.263 \pm 0.185$
finest	NoCD	7	-3.523 ± 0.030	-0.020 ± 0.029
finest	correct	7	-3.503 ± 0.035	—
$\{B\}, \{DE\}$	NoBC	3	-1.245 ± 0.065	-0.001 ± 0.001
$\{B\}, \{DE\}$	NoCD	3	-1.244 ± 0.066	$+0.000 \pm 0.000$
$\{B\}, \{DE\}$	correct	3	-1.244 ± 0.066	—

6.3 Robustness: two damage regimes

We separate the two tiers of Remark 4.

Tier-1 (prior bias) — CompleteGraph (Table 3). Deleting the inter-cluster edge $B \rightarrow C$ (NoBC) or the edge $C \rightarrow D$ (NoCD) leaves all identifiability verdicts intact but corrupts the priors of the affected arms. The damage is real but partially self-healing: under NoBC the fine-grained method’s paired per-seed gap is $+0.263 \pm 0.185$ — a modest mean shift hiding a six-fold variance inflation (± 0.199 vs. ± 0.035 s.e.), i.e. some seeds recover fully and some stay trapped; NoCD is absorbed entirely (-0.020 ± 0.029). The coarse partition is unaffected by either (± 0.001), sitting at its Prop. 4 floor: under $\{B\}, \{D, E\}$ both misspecified edges either lie inside a cluster or are made redundant at the quotient level ($C \rightarrow E$ keeps the $\{C\} \rightarrow \{D, E\}$ cluster edge alive when $C \rightarrow D$ is deleted).

Tier-2 (arm damage) — LightTunnel (Table 4). The misspecified DAG WrongBG adds a single spurious intra-cluster edge $B \rightarrow G$. Together with the (real) latent confounding $B \leftrightarrow G$ this forms a bow, so $P(Y \mid (B))$ is not identifiable from the assumed fine graph: the identity-partition exploration set provably loses its best arm $\{B\}$ (ground-truth arm values: $\min_b \mathbb{E}[Y \mid (B=b)] = 0.40$ vs. 0.60 for the best surviving singleton $\{G\}$). No amount of interventional data restores a deleted arm, and the experiment shows exactly that: the paired gap is $+0.195 \pm 0.000$ at every seed. Under the coarse partition $\{R, G, B\}, \{P_1, P_2\}$ the bow is intra-cluster: Prop. 2 applies, and the CCBO trajectories under the correct and misspecified DAGs are *byte-for-byte identical* at all 10 seeds (the cluster arm also attains $\mathbb{E}[Y \mid (R, G, B=0)] \approx 0$, dominating every singleton).

7 Discussion

- Prop. 2 is structural, not statistical: coarsening acts as a hard regularizer on the hypothesis space, trading resolution (Prop. 4) for immunity to an entire class of misspecifications. The practitioner chooses where on this dial to sit by choosing how much structure to commit to.

Table 4: Robustness under misspecification (LightTunnel; 10 seeds). “Gap” is the paired per-seed difference to the same partition under the correct DAG; ± 0.000 for the coarse partition reflects byte-identical trajectories (Prop. 2).

Partition	Assumed DAG	$ \mathcal{S}^\pi $	Final Y	Gap vs. correct
finest	WrongBG	4	0.597 ± 0.000	$+0.195 \pm 0.000$
finest	correct	5	0.402 ± 0.000	—
{RGB},{P1P2}	WrongBG	2	0.046 ± 0.007	$+0.000 \pm 0.000$
{RGB},{P1P2}	correct	2	0.046 ± 0.007	—

- The damage taxonomy (Remark 4) reframes “is CBO robust to a wrong graph?”: prior bias is self-healing under any reasonable acquisition; arm damage is not. Robustness analyses that only perturb priors miss the failure mode that matters.
- The partition need not be guessed once and forever: Prop. 3 makes refinement monotonically safe (identifiable arms are never lost), although validity must be re-checked (Remark 3). Discovering the partition online from interventional data — e.g. with the RePaRe routine of Madaleno et al. [2026] — is a natural recursive extension that we develop in follow-up work.

8 Limitations

(i) The gated design (Eq. 3) commits to the assumed structure; the un-gated variant (explore non-identified arms under an uninformative prior) converts arm damage into exploration overhead and deserves a systematic comparison. (ii) The GP estimator cascade behind Eq. (4) is sound but not complete (Assumption 2). (iii) Three benchmarks, one physical simulator; no real-chamber hardware runs yet. (iv) Faithfulness does not transfer to the cluster level [Wahl et al., 2024], so cluster-level conditional independences observed in data must not be read back as fine-grained structure. (v) We evaluate at unit intervention costs; cost-asymmetric settings change where coarsening pays.

9 Conclusion

CCBO makes causal Bayesian optimization usable at the level of structural knowledge practitioners actually have — clusters — with a complete identification gate, a provable invariance to intra-cluster misspecification, and an exact characterization of what coarseness costs. On the spectrum from structure-free BO to oracle-graph CBO, it provides the missing, principled middle; refining the partition as evidence accumulates closes the remaining gap and is the subject of ongoing work.

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